

Lab_Assignment_7.1

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Batch 17

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Task Description #1 (Syntax Errors – Missing Parentheses in Print Statement)

Task: Provide a Python snippet with a missing parenthesis in a print statement (e.g., `print "Hello"`). Use AI to detect and fix the syntax error.

Bug: Missing parentheses in print statement

```
def greet():
```

```
print "Hello, AI Debugging Lab!"
```

```
greet()
```

Requirements:

- Run the given code to observe the error.
- Apply AI suggestions to correct the syntax.
- Use at least 3 assert test cases to confirm the corrected code works.

Expected Output #1:

- Corrected code with proper syntax and AI explanation.

Prompt :

#correct the indentation error in the code above.

#add missing parentheses in the print statement.

Code :

```
def greet():
```

```
print "Hello, AI Debugging Lab!"
```

```
greet()
```

#correct the indentation error in the code above.

#add missing parentheses in the print statement.

```
def greet():  
    print ("Hello, AI Debugging Lab!")  
greet()
```

Output

Hello, AI Debugging Lab!

Task Description #2 (Incorrect condition in an If Statement)

Task: Supply a function where an if-condition mistakenly uses = instead of ==. Let AI identify and fix the issue.

Bug: Using assignment (=) instead of comparison (==)

```
def check_number(n):  
    if n = 10:  
        return "Ten"  
    else:  
        return "Not Ten"
```

Requirements:

- Ask AI to explain why this causes a bug.
- Correct the code and verify with 3 assert test cases.

Expected Output #2:

- Corrected code using == with explanation and successful test execution.

Prompt

Bug: Using assignment (=) instead of comparison (==)

Code

Bug: Using assignment (=) instead of comparison (==)

```
def is_even(num):  
    if num == 0:  
        return True  
    elif num % 2 == 0:  
        return True
```

```
else:  
    return False
```

Output

Corrected code using == with explanation and successful test

Task Description #3 (Runtime Error – File Not Found)

Task: Provide code that attempts to open a non-existent file and crashes. Use AI to apply safe error handling.

Bug: Program crashes if file is missing

```
def read_file(filename):  
    with open(filename, 'r') as f:  
        return f.read()  
print(read_file("nonexistent.txt"))
```

Requirements:

- Implement a try-except block suggested by AI.
- Add a user-friendly error message.
- Test with at least 3 scenarios: file exists, file missing, invalid path.

Expected Output #3:

- Safe file handling with exception management.

Prompt

```
#Implement a try-except block  
#Test with at least 3 scenarios: file exists, file missing, invalid path.
```

Code

```
#Implement a try-except block  
#Test with at least 3 scenarios: file exists, file missing, invalid path.  
def read_file(filename):  
    try:  
        with open(filename, 'r') as f:  
            return f.read()
```

```
except FileNotFoundError:
    return "File not found."
except Exception as e:
    return f"An error occurred: {e}"
```

Output

- Safe file handling with exception management.

Task Description #4 (Calling a Non-Existent Method)

Task: Give a class where a non-existent method is called (e.g., `obj.undefined_method()`). Use AI to debug and fix.

Bug: Calling an undefined method

```
class Car:
    def start(self):
        return "Car started"
my_car = Car()
print(my_car.drive()) # drive() is not defined
```

Requirements:

- Students must analyze whether to define the missing method or correct the method call.
- Use 3 assert tests to confirm the corrected class works.

Expected Output #4:

- Corrected class with clear AI explanation.

Prompt

```
#define the drive() method in the Car class to fix the AttributeError.
# drive() is not defined
```

Code

```
#define the drive() method in the Car class to fix the AttributeError.
# drive() is not defined
class Car:
```

```
def start(self):  
    return "Car started"  
  
def drive(self):  
    return "Car is driving"  
  
my_car = Car()  
print(my_car.drive())
```

Output

Car is driving

Task Description #5 (TypeError – Mixing Strings and Integers in Addition)

Task: Provide code that adds an integer and string ("5" + 2) causing a TypeError. Use AI to resolve the bug.

Bug: TypeError due to mixing string and integer

```
def add_five(value):  
    return value + 5  
  
print(add_five("10"))
```

Requirements:

- Ask AI for two solutions: type casting and string concatenation.
- Validate with 3 assert test cases.

Expected Output #5:

Prompt

```
# Implement type checking to handle the TypeError when adding an integer to a string.  
#Corrected code that runs successfully for multiple inputs.
```

Code

```
# Implement type checking to handle the TypeError when adding an integer to a string.  
#Corrected code that runs successfully for multiple inputs.  
  
def add_five(value):
```

```
if isinstance(value, (int, float)):
    return value + 5
else:
    return "Input must be a number."
print(add_five(10)) # Output: 15
print(add_five(3.5)) # Output: 8.5
print(add_five("10")) # Output: Input must be a number.
```

Output

```
15
8.5
Input must be a number.
```